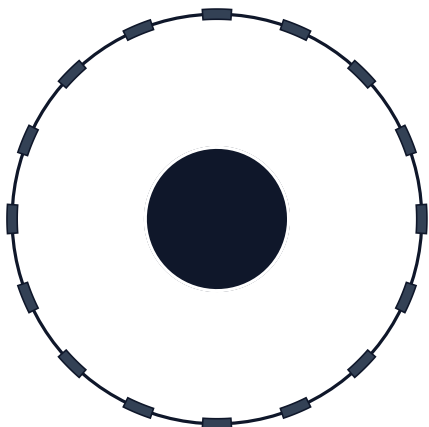
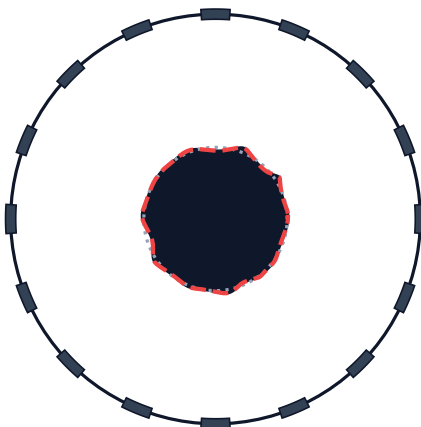


Surrogate-Assisted Levenberg--Marquardt Inversion: Targets 1--3 (Convex Morphologies)

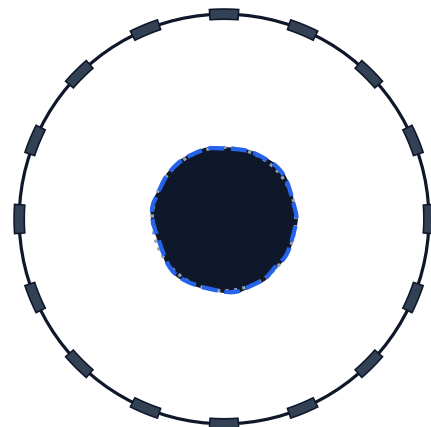
Target #1: Circle (Convex)
[Ground Truth]



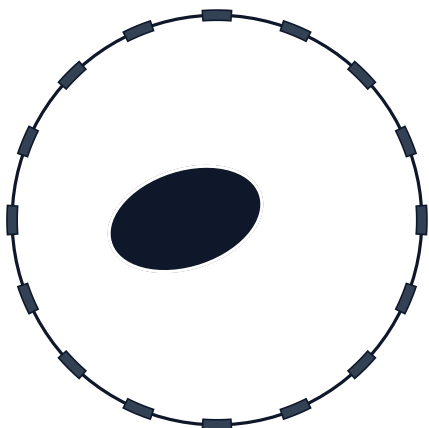
Deterministic Inversion
IoU = 0.9615 | RMS = 9.3 mm



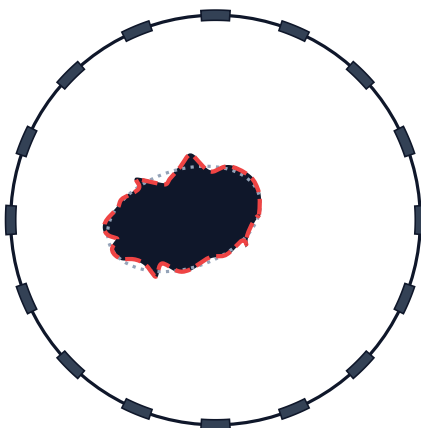
Ensemble Mean
IoU = 0.9765 | RMS = 6.0 mm



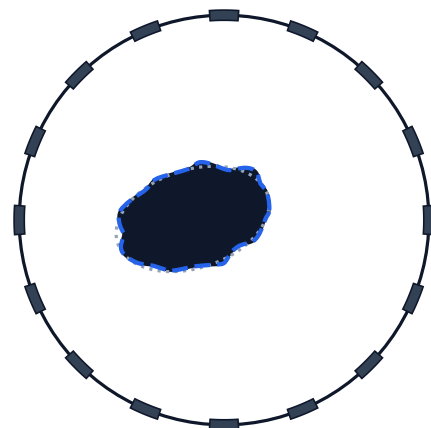
Target #2: Ellipse (Convex)
[Ground Truth]



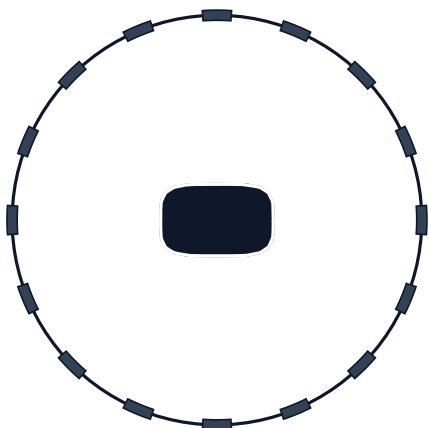
Deterministic Inversion
IoU = 0.9034 | RMS = 18.7 mm



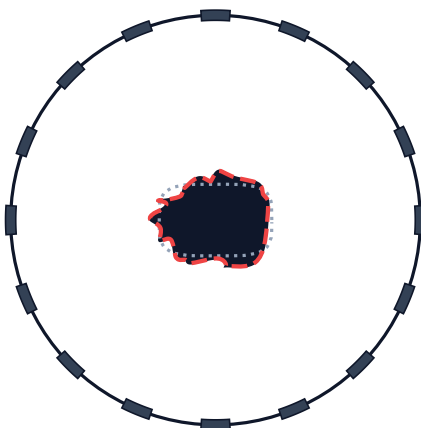
Ensemble Mean
IoU = 0.9377 | RMS = 12.5 mm



Target #3: Rectangle (Convex)
[Ground Truth]



Deterministic Inversion
IoU = 0.7846 | RMS = 28.1 mm



Ensemble Mean
IoU = 0.6610 | RMS = 70.0 mm

